Control work 4 · Numerical inequalities

The Chapter II assessment on properties, adding, multiplying and powers — and the mistakes lesson that follows.

LESSON 47–48 | 2 × 40 MIN | ALGEBRA 8, §§11–14 | STAGE 9 · 4.3

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess the definition, the four properties, adding, multiplying and powers.
- Sort every lost mark into one of four error types.
- Re-solve every task that was lost.

TEXTBOOK

National	Revision of §§11–14
Cambridge	Learner's Book pp. 96–102
Workbook	Workbook 4.3

01 Explanation

Lesson 47 — the paper (40 minutes)

Two variants of seven tasks, 2 marks each:

- task 1 · compare two numbers by the difference (§11)
- task 2 · a property applied to $a > b$ (§12)
- task 3 · solve a one-step inequality with a negative coefficient (§12)
- tasks 4–5 · estimate a sum, difference, product or quotient (§13)
- task 6 · estimate a square or a root (§14)
- task 7 · prove an inequality using a square (§11)

Lesson 48 — work on mistakes (40 minutes)

THE FOUR ERRORS THIS PAPER PRODUCES

1. **The sign not reversed** after dividing by a negative number.
2. **Inequalities subtracted or divided term by term.**
3. **An interval containing zero squared end to end**, losing the value 0.
4. **The positivity condition ignored** when multiplying or squaring.

The Hard set below is seven pieces of wrong working drawn from these four.

02 Worked examples

EXAMPLE 1 **Find the error: $-2x > 8$, so $x > -4$**

1. Dividing by -2 — a negative number.
   Property 4 applies.

2. The sign must reverse.

3. $x < -4$
   Check: $x = -5$ gives $10 > 8$ ✓

ANSWER

$x < -4$

EXAMPLE 2 **Find the error: $-3 < a < 2$, so $9 < a^2 < 4$**

1. The written range is impossible — 9 is not less than 4.
   That alone signals an error.

2. The interval contains 0, so the ends may not be squared directly.

3. $a^2 = 0$ at $a = 0$; the largest comes from -3.

4. $0 \le a^2 < 9$

ANSWER

$0 \le a^2 < 9$

03 Interactive model

Lesson 48: show each piece of wrong working, vote on the error type, then reveal.

Diagnose the error

Claimed: $-2x > 8 \Longrightarrow x > -4$

1. Dividing by -2 is dividing by a negative number.

2. So the inequality sign turns: $x < -4$.

3. Test $x = -5$: $-2(-5) = 10 > 8$ ✓, and $-5 < -4$ ✓.

The test settles it.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Estimate a range	Oraliqni baholash	Оценить промежуток
Reverse the sign	Ishorani almashtirish	Изменить знак неравенства
Counter-example	Qarshi misol	Контрпример
Prove	Isbotlash	Доказать
Condition	Shart	Условие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$-5x \geq 20$ gives:

- $x \geq -4$
- $x \leq -4$
- $x \geq 4$
- $x \leq 4$

From $a > b$ and $c > d$, which is guaranteed?

- $a - c > b - d$
- $a + c > b + d$
- $ac > bd$
- $\frac{a}{c} > \frac{b}{d}$

If $-2 < a < 5$ then:

- $4 < a^2 < 25$
- $0 \leq a^2 < 25$
- $-4 < a^2 < 25$
- $0 < a^2 < 4$

To prove $a^2 + 1 \geq 2a$ you should:

- try a = 3
- take the difference and factorise
- divide both sides by a
- square both sides

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Compare* $\frac{2}{5}$ *and* $\frac{3}{8}$ *by the difference*

   ANSWER $\frac{2}{5} > \frac{3}{8}$

2. ***Task 2.*** *$a > b$. Compare $a + 6$ and $b + 6$*

   ANSWER $a + 6 > b + 6$

3. ***Task 3.*** *Solve $-2x > 8$*

   ANSWER $x < -4$

4. ***Task 4.*** *$1 < a < 4$ and $2 < b < 5$. Find the range of $a + b$*

   ANSWER $3 < a + b < 9$

5. ***Task 5.*** *Same ranges: find the range of ab*

   ANSWER $2 < ab < 20$

6. ***Task 6.*** *$2 < a < 5$. Find the range of a^2*

   ANSWER $4 < a^2 < 25$

7. ***Task 7.*** *Prove $a^2 + 1 \geq 2a$*

   ANSWER The difference is $(a - 1)^2 \geq 0$.

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Compare* $\frac{4}{7}$ *and* $\frac{5}{9}$ *by the difference*

   ANSWER $\frac{4}{7} > \frac{5}{9}$

2. ***Task 2.*** $a > b$. *Compare* $-3a$ *and* $-3b$

   ANSWER $-3a < -3b$

3. ***Task 3.*** *Solve* $-5x + 1 \leq 11$

   ANSWER $x \geq -2$

4. ***Task 4.*** $2 < a < 5$ *and* $1 < b < 3$. *Find the range of* $a - b$

   ANSWER $-1 < a - b < 4$

5. ***Task 5.*** *Same ranges: find the range of* $\frac{a}{b}$

   ANSWER $\frac{2}{3} < \frac{a}{b} < 5$

6. ***Task 6.*** $9 < a < 36$. *Find the range of* $\sqrt{a}$

   ANSWER $3 < \sqrt{a} < 6$

7. ***Task 7.*** *Prove* $a^2 + b^2 \geq 2ab$

   ANSWER The difference is $(a - b)^2 \geq 0$.

Hard · stretch and reason

7 PROBLEMS

1. *Find the error:* $-2x > 8$, *so* $x > -4$

   ANSWER The sign was not reversed. Correct: $x < -4$.

2. *Find the error: from* $5 > 3$ *and* $10 > 1$, $5 - 10 > 3 - 1$

   ANSWER Inequalities cannot be subtracted. Reverse the second, then add.

3. *Find the error:* $-3 < a < 2$, *so* $9 < a^2 < 4$

   ANSWER The interval contains 0. Correct: $0 \le a^2 < 9$.

4. *Find the error: from* $2 > -5$ *and* $3 > -4$, $6 > 20$

   ANSWER Multiplication needs all terms positive; here $6 < 20$.

5. *Find the error:* $a > b$, *so* $a^2 > b^2$

   ANSWER Needs $a > b > 0$. Counter-example $-2 > -5$ but $4 < 25$.

6. *Find the error:* $1 < b < 4$, *so* $1 < \frac{1}{b} < 4$

   ANSWER Reciprocals reverse the order. Correct: $\frac{1}{4} < \frac{1}{b} < 1$.

7. *Find the error:* “$a^2 \ge 2a$ *is true for all* a *because* $(a - 1)^2 \ge 0$”

   ANSWER The difference of a^2 and $2a$ is $a^2 - 2a = a(a - 2)$, not a square; the claim is false for $a = 1$.

07 Homework

After the work-on-mistakes lesson

Re-solve every task you lost marks on. Quarter III opens with inequalities in one unknown.

1. *Solve* $-4x + 5 > 21$
2. $3 < a < 8$ *and* $2 < b < 4$. *Find the ranges of* $a + b$, $a - b$, ab *and* $\frac{a}{b}$

3. $-5 < a < 1$. *Find the range of* a^2
4. Write out the four errors from this lesson in your own words with an example each.